0.1 Riemann-Stieltjes Integral

Definition 0.1.1. A finite subset $P \subset [a,b]$ containing a, b is called a *Partition* of $[a,b]$.
Conventionally, denote $P = \{x_0, x_1, \dots, x_n\}$ where $a = x_0 < x_1 < \cdots < x_n = b$.
The set of Partitions of $[a,b]$ is denoted as $\mathcal{P}[a,b]$.

Definition 0.1.2. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a,b] \to \mathbb{R}$ be a monotone-increasing.
Let a partition $P \in \mathcal{P}[a,b]$ be given, and denote $P = \{x_0, x_1, \dots, x_n\}$. For each $1 \le i \le n$, define

$$\Delta\alpha_i \overset{\text{def}}{=} \alpha(x_i) - \alpha(x_{i-1}) \underset{\alpha \text{ mono.}}{\ge} 0$$

and

$$M_i \overset{\text{def}}{=} \sup\{f(t) \mid x_{i-1} \le t \le x_i\} \text{ and } m_i \overset{\text{def}}{=} \inf\{f(t) \mid x_{i-1} \le t \le x_i\}$$

And, define *Upper Sum* and *Lower Sum* as:

$$U(f,P,\alpha) \overset{\text{def}}{=} \sum_{i=1}^{n} M_i \cdot \Delta\alpha_i \text{ and } L(f,P,\alpha) \overset{\text{def}}{=} \sum_{i=1}^{n} m_i \cdot \Delta\alpha_i$$

Now, define *Upper integral* and *Lower integral* as:

$$\overline{\int_a^b} f d\alpha \overset{\text{def}}{=} \inf_{P \in \mathcal{P}[a,b]} U(f,P,\alpha) \text{ and } \underline{\int_a^b} f d\alpha \overset{\text{def}}{=} \sup_{P \in \mathcal{P}[a,b]} L(f,P,\alpha)$$

We say that f is *Riemann-Stieltjes Integrable* with respect to α if:

$$\overline{\int_a^b} f d\alpha = \underline{\int_a^b} f d\alpha$$

The set of Riemann-Stieltjes Integrable functions with respect to α as $\mathcal{R}(\alpha)$.

Lemma 0.1.3. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a,b] \to \mathbb{R}$ be a monotone-increasing.
For any partitions $P, Q \subset [a,b]$, and a refinement $P^* \supset P$,

$$L(f,P,\alpha) \le L(f,P^*,\alpha) \le \underline{\int_a^b} f d\alpha \le \overline{\int_a^b} f d\alpha \le U(f,P^*,\alpha) \le U(f,P,\alpha)$$

and

$$L(f,P,\alpha) \le U(f,Q,\alpha)$$

0.1.1 Properties of Riemann-Stieltjes Integral

Theorem 0.1.4. Let $f : [a, b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \to \mathbb{R}$ be monotone-increasing. TFAE:

1. f is Riemann-Stieltjes Integrable with respect to α on $[a, b]$.
2. For any $\varepsilon > 0$, there exists a Partition $P \in \mathcal{P}[a, b]$ such that $U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon$.
3. For some $A \in \mathbb{R}$, for any $\varepsilon > 0$, there exists $P_0 \in \mathcal{P}[a, b]$ such that

$$\forall P \in \mathcal{P}[a, b],\ \ P \supseteq P_0 \implies |S(f, P, \alpha) - A| < \varepsilon$$

where Riemann-Stieltjes Sum $S(f, P, \alpha) \overset{\text{def}}{=} \sum_{i=1}^{n} f(t_i)[\alpha(x_i) - \alpha(x_{i-1})]$, for some $t_i \in [x_{i-1}, x_i]$.

Lemma 0.1.5. Let $f : [a, b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \to \mathbb{R}$ be a monotone-increasing.
If $f, g \in \mathcal{R}(\alpha)$ and $c \in \mathbb{R}$, then $f + g,\ \ cf \in \mathcal{R}(\alpha)$ with

$$\int_a^b (f+g)d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$$
$$\int_a^b (cf)d\alpha = c\int_a^b f d\alpha$$

Lemma 0.1.6. Let $f : [a, b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a, b] \to \mathbb{R}$ be a monotone-increasing, and $a < c < b$.

f is Stieltjes Integrable on $[a, b]$ *if and only if* f is Stieltjes Integrable on $[a, c]$ and $[c, b]$

In this case, the following equality holds:

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha$$

Lemma 0.1.7. Let $f : [a, b] \to \mathbb{R}$ be a bounded map, and $\alpha, \beta : [a, b] \to \mathbb{R}$ be monotone-increasing.
If $f \in \mathcal{R}(\alpha)$, $f \in \mathcal{R}(\beta)$ and $c > 0$, then $f \in \mathcal{R}(\alpha + \beta)$ and $f \in \mathcal{R}(c\alpha)$ and $|f| \in \mathcal{R}(\alpha)$ with

$$\int_a^b f d(\alpha+\beta) = \int_a^b f d\alpha + \int_a^b f d\beta$$
$$\int_a^b f d(c\alpha) = c\int_a^b f d\alpha$$
$$\left|\int_a^b f d\alpha\right| \leq \int_a^b |f| d\alpha$$

0.1.2 Riemann-Stieltjes Integrable

> **Theorem 0.1.8.** If $f : [a, b] \to \mathbb{R}$ is continuous, then $f \in \mathcal{R}(\alpha)$ for any monotone increasing $\alpha : [a, b] \to \mathbb{R}$.

> **Theorem 0.1.9.** If $f : [a, b] \to \mathbb{R}$ is monotone and $\alpha : [a, b] \to \mathbb{R}$ is monotone increasing continuous, then $f \in \mathcal{R}(\alpha)$.

> **Theorem 0.1.10.** If $f : [a, b] \to \mathbb{R}$ is map with $m \leq f(t) \leq M$ for all $t \in [a, b]$ and ϕ is continuous on $[m, M]$, then $\phi \circ f$ is Riemann-Stieltjes integrable with respect to α on $[a, b]$

> **Theorem 0.1.11.** If $\alpha : [a, b] \to \mathbb{R}$ is continuously differentiable monotone increasing, then
>
> $$f \in \mathcal{R}(\alpha) \text{ if and only if } f\alpha' \in \mathcal{R}(x)$$
>
> In this case,
>
> $$\int_a^b f(x)d\alpha(x) = \int_a^b f(x)\alpha'(x)dx$$

0.1.3 Riemann-Integral

Definition 0.1.12. Let $\mathcal{P}[a,b]$ be a set of partitions. Define a *Norm* as:

$$\|\cdot\| : \mathcal{P}[a,b] \to \mathbb{R} : P \mapsto \max\{|x_i - x_{i-1}| \mid i = 1, 2, \dots, n\} \text{ where } P = \{x_0, x_1, \dots, x_n\}$$

Theorem 0.1.13. Let $f : [a,b] \to \mathbb{R}$ and $A \in \mathbb{R}$, TFAE:

1. f is Riemann-Integral with $\displaystyle\int_a^b f = A$.
2. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $\forall P \in \mathcal{P}[a,b],\ \ \|P\| < \delta \implies |R(f,P) - A| < \varepsilon$.
3. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $\forall P \in \mathcal{P}[a,b],\ \ P \supseteq P_0 \implies |R(f,P) - A| < \varepsilon$.

The statement 2) can be written as: $\displaystyle\lim_{\|P\| \to 0} R(f,P) = A$.

Theorem 0.1.14. Let $f : [a, b] \to \mathbb{R}$ be a Riemann-Integrable function. Define

$$F : [a, b] \to \mathbb{R} : x \mapsto \int_a^x f(t)dt$$

Then, F is Continuous on $[a, b]$. Particular, if f is continuoous at $x_0 \in [a, b]$, then F is differentiable at x_0 with

$$F'(x_0) = f(x_0)$$

Theorem 0.1.15. The fundamental theorem of calculus
If $f : [a, b] \to \mathbb{R}$ is Riemann-Integrable and there exists differetiable function $F : [a, b] \to \mathbb{R}$ such that $F' = f$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

0.1.4 Bounded Variation Function

Definition 0.1.16. Let $f : [a, b] \to \mathbb{R}$ be a bounded function and $P = \{x_0, x_1, \ldots, x_n\} \in \mathcal{P}[a, b]$.
Define a *Variation* of f with respects to P as:

$$V_a^b(f, P) \overset{\text{def}}{=} \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})|$$

If the set $\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$ is bounded above, then f is called *Bounded Variation Function*.
In this case, define a *Total Variation* of f as:

$$TV_a^b(f) \overset{\text{def}}{=} \sup\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$$

In simply, denote $V(f, P) = V_a^b(f, P)$ and $TV(f) = TV_a^b(f)$.

Lemma 0.1.17. Let $f, g : [a, b] \to \mathbb{R}$ be Bounded Variation Functions, $\alpha \in \mathbb{R}$ be a real number.
Then, $f + g, fg, \alpha f$ are Bounded Variation Functions and satisfy

$$\begin{aligned} &TV(f+g) \le TV(f) + TV(g) \\ &TV(fg) \le \sup_{t \in [a,b]} |g(t)| \cdot TV(f) + \sup_{t \in [a,b]} |f(t)| \cdot TV(g) \\ &TV(\alpha f) = |\alpha| \cdot TV(f) \end{aligned}$$

Lemma 0.1.18. Let $f : [a, b] \to \mathbb{R}$ be a map, and $a < c < b$.

f is bounded variation on $[a, b]$ *if and only if* f is bounded variation on $[a, c]$ and $[c, b]$

In this case, the following equality holds:

$$TV_a^c(f) + TV_c^b(f) = TV_a^b(f)$$

Theorem 0.1.19. Let $f : [a, b] \to \mathbb{R}$ be given.

f is bouned varation *if and only if* For some monotone increasing $g, h : [a, b] \to \mathbb{R}$, $f = g - h$

Proposition 0.1.20. Let $f : [a,b] \to \mathbb{R}$ be a continuously differentiable function.
If f is bounded variation, then

$$\int_a^b |f'(x)|dx \leq TV_a^b(f)$$

Proof. Let $P = \{x_0, \dots, x_n\}$ be a Partition on $[a,b]$. By the Mean-Value Theorem, for each $1 \leq i \leq n$, for some $t_i \in [x_{i-1}, x_i]$,

$$V_a^b(f,P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i$$

Meanwhile,

$$L(|f'|,P) \leq \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq U(|f'|,P)$$

Both Lower Sum and Upper Sum converge to the $\displaystyle\int_a^b |f'(x)|dx$, we can complete this conversation.
Given $\varepsilon > 0$, there exists a partition P on $[a,b]$ such that

$$\int_a^b |f'(x)|dx \leq V_a^b(f,P) + \varepsilon \leq TV_a^b(f) + \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, therefore we obtain the result. $\square$

Note: if f is continuously differentiable, then

For any $\varepsilon > 0$, there exists a partition P on $[a,b]$ such that $\left|\displaystyle\int_a^b |f'(x)|dx - V_a^b(f,P)\right| < \varepsilon$.

And, we can weaken the condition: continuously differentiable $\to$ differentiable and the derivative is Riemann-Integrable.

Proposition 0.1.21. If $f : [a,b] \to \mathbb{R}$ is differentiable and the derivative f' is bounded,
then f is bounded variation.

Proof. For any partition P on $[a,b]$, by the Mean Value Theorem, there exists $t_i \in (x_{i-1}, x_i)$ such that

$$V(f,P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n \sup|f'| \cdot \Delta x_i = \sup|f'| \cdot 1$$

Since P was arbitrary, f is bounded variation. $\square$

Definition 0.1.22. Let $\alpha : [a, b] \to \mathbb{R}$ be a bounded variation.
A map f is said to be *Riemann-Stieltjes integrable* with respect to α if:

for some monotone increasing α_1, α_2, $\alpha = \alpha_1 - \alpha_2$ such that $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$

and in this case, define

$$\int_a^b f d\alpha = \int_a^b f d\alpha_1 - \int_a^b f d\alpha_2$$

Lemma 0.1.23. If $f : [a, b] \to \mathbb{R}$ is continuous, then $f \in \mathcal{R}(\alpha)$ for any bounded varation α.

Theorem 0.1.24. Let $f, \alpha : [a, b] \to \mathbb{R}$ be bounded variation functions. If $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ with

$$\int_a^b f d\alpha + \int_a^b \alpha df = f(b)\alpha(b) - f(a)\alpha(a)$$

Corollary 0.1.25. If $f : [a, b] \to \mathbb{R}$ is monotone increasing and $g : [a, b] \to \mathbb{R}$ is continuous.
Then, there exists $c \in [a, b]$ such that

$$\int_a^b f(x)g(x)dx = f(a)\int_a^c g(x)dx + f(b)\int_c^b g(x)dx$$

0.1.5 Step function

Definition 0.1.26. Define a *Step function* as:

$$I : \mathbb{R} \to \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x \le 0 \\ 1 & \text{if } x > 0 \end{cases}$$

Lemma 0.1.27. If $a < s < b$, f is bounded map on $[a, b]$, f is continuous at s, and $\alpha(x) = I(x - s)$, then

$$\int_a^b f d\alpha = f(s)$$

Theorem 0.1.28. Suppose that $c_n \ge 0$ and $\sum c_n$ converges. And, $\{s_n\}$ is a sequence of distinct points in (a, b). Define

$$\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$$

If f is Continuous on $[a, b]$, then

$$\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$$